Windows Security Log Analysis

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1. Objective

The purpose of this exercise was to explore Windows Security logs and understand how authentication, privilege assignment, and credential-related activities are recorded within Windows. The investigation focused on commonly observed security events and the information they provide during security monitoring and incident investigations.

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2. Lab Environment

Component	Details
Operating System	Windows 11
Log Source	Windows Security Log
Analysis Tool	Event Viewer

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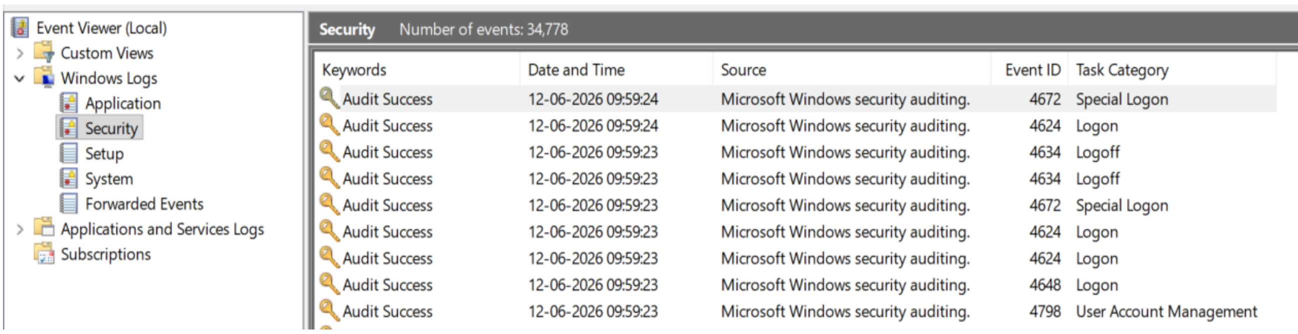
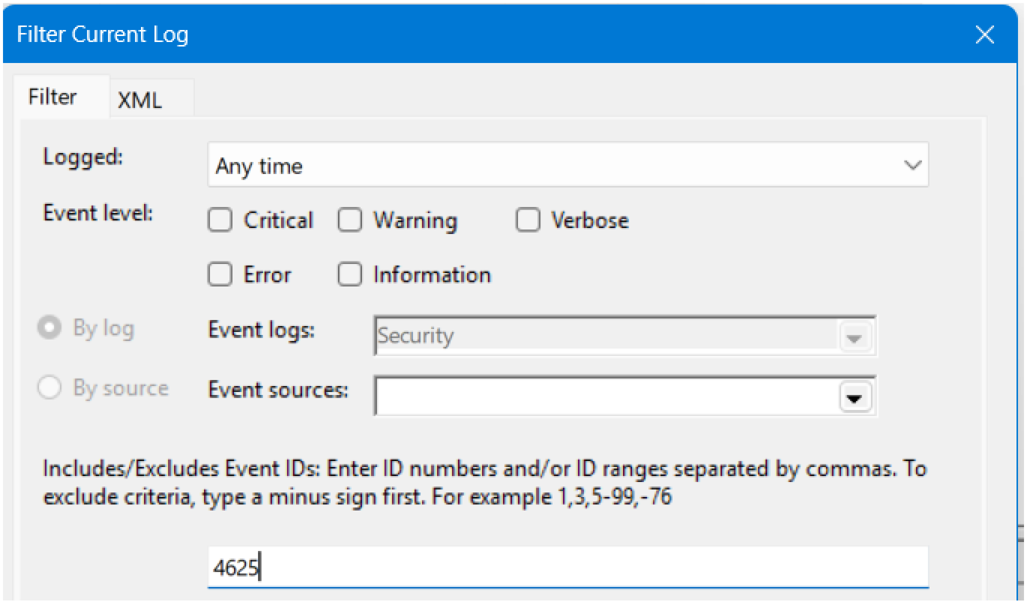
3. Accessing Security Logs

Windows Security logs were accessed through Event Viewer using the following path:

Windows Logs
└── **Security**



The Security log contained a large number of events generated by normal system activity. To simplify the investigation, the **Filter Current Log** feature was used to isolate specific Event IDs.

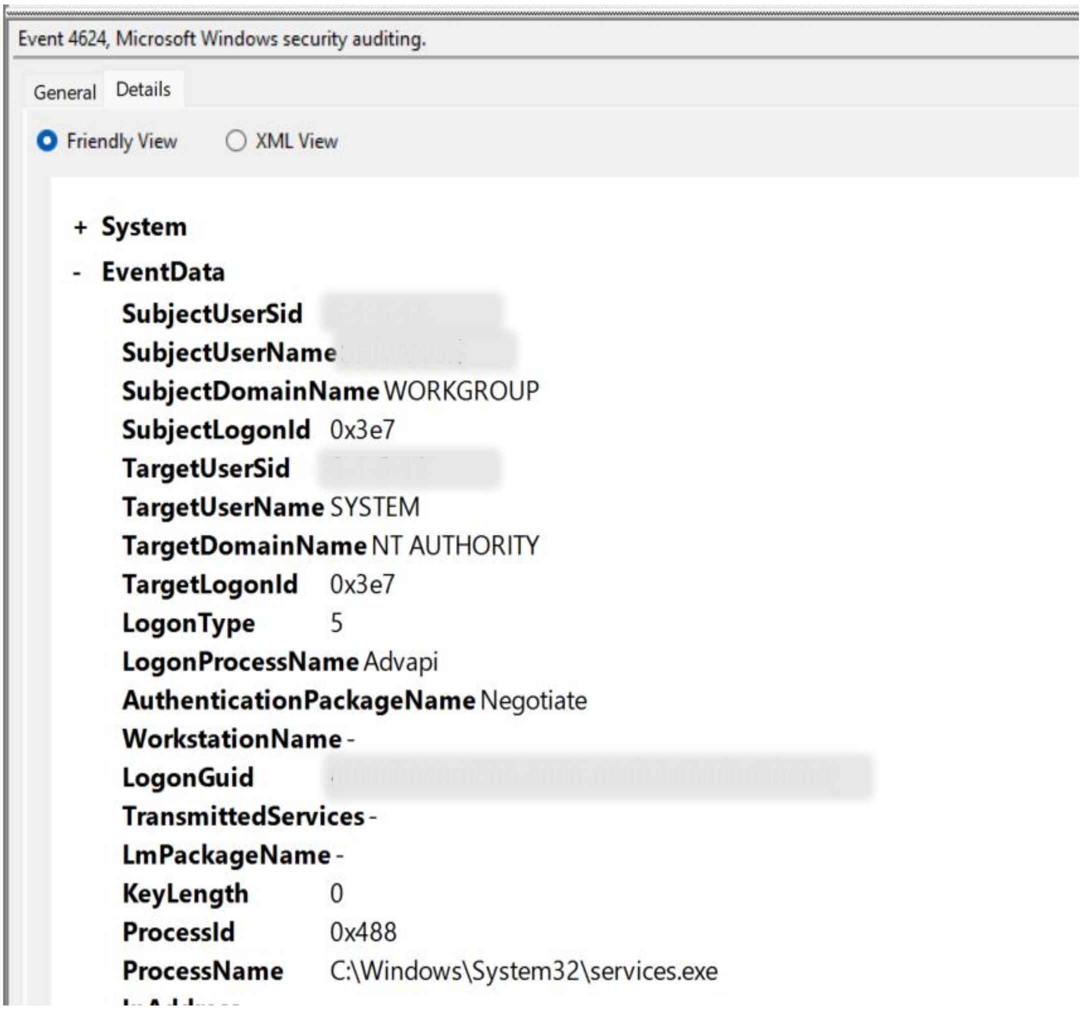


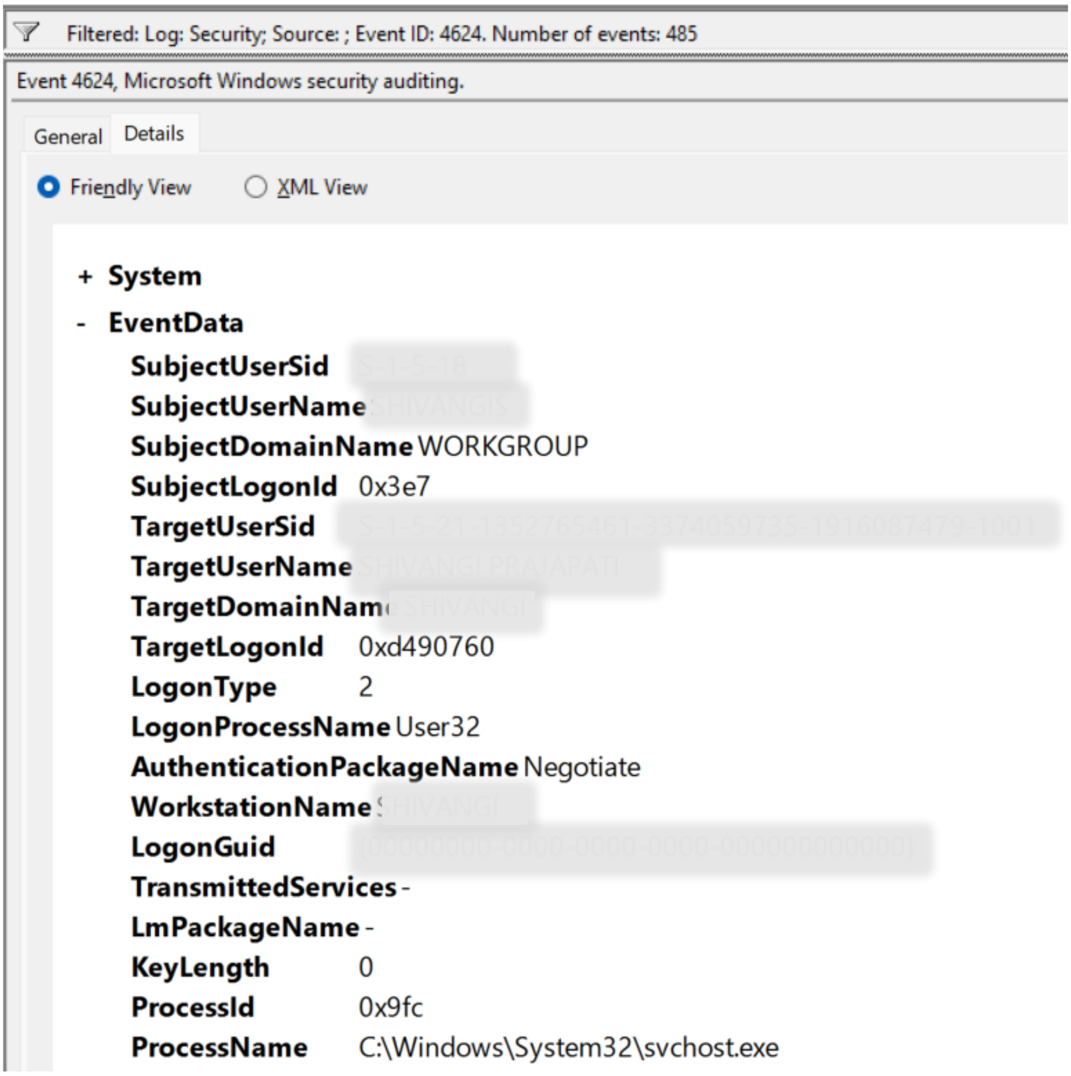
Observation

Filtering reduced the number of visible events and allowed specific security-related activities to be investigated individually.

4. Event ID 4624 – Successful Logon

Event ID 4624 records successful authentication activity and is one of the most commonly reviewed security events during investigations.





Observation

Multiple Event ID 4624 entries were identified. These events indicate successful authentication attempts performed by users, services, or system processes.

Key Fields Observed

- Security Identifier (SID)
- Account Name
- Account Domain
- Logon Type
- Workstation Name
- Process Name
- Authentication Package

Analysis

The event details provide visibility into who authenticated, how the authentication occurred, and which process initiated the request. Successful logon events are frequently used to establish timelines during investigations and verify account activity.

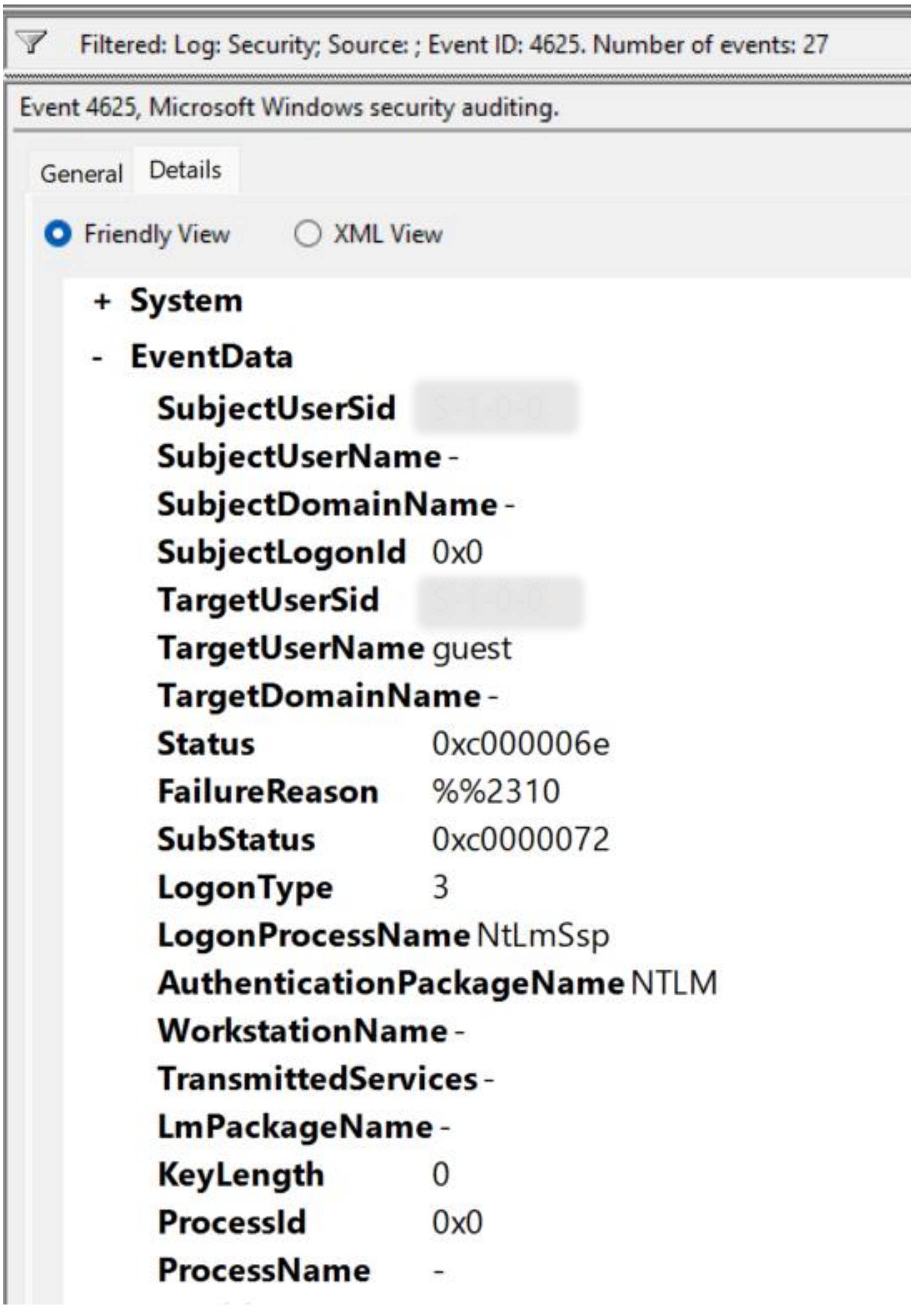
Logon Type values are particularly important because they reveal how access was obtained. Interactive logons indicate local user activity, while network or remote logons may require additional validation.

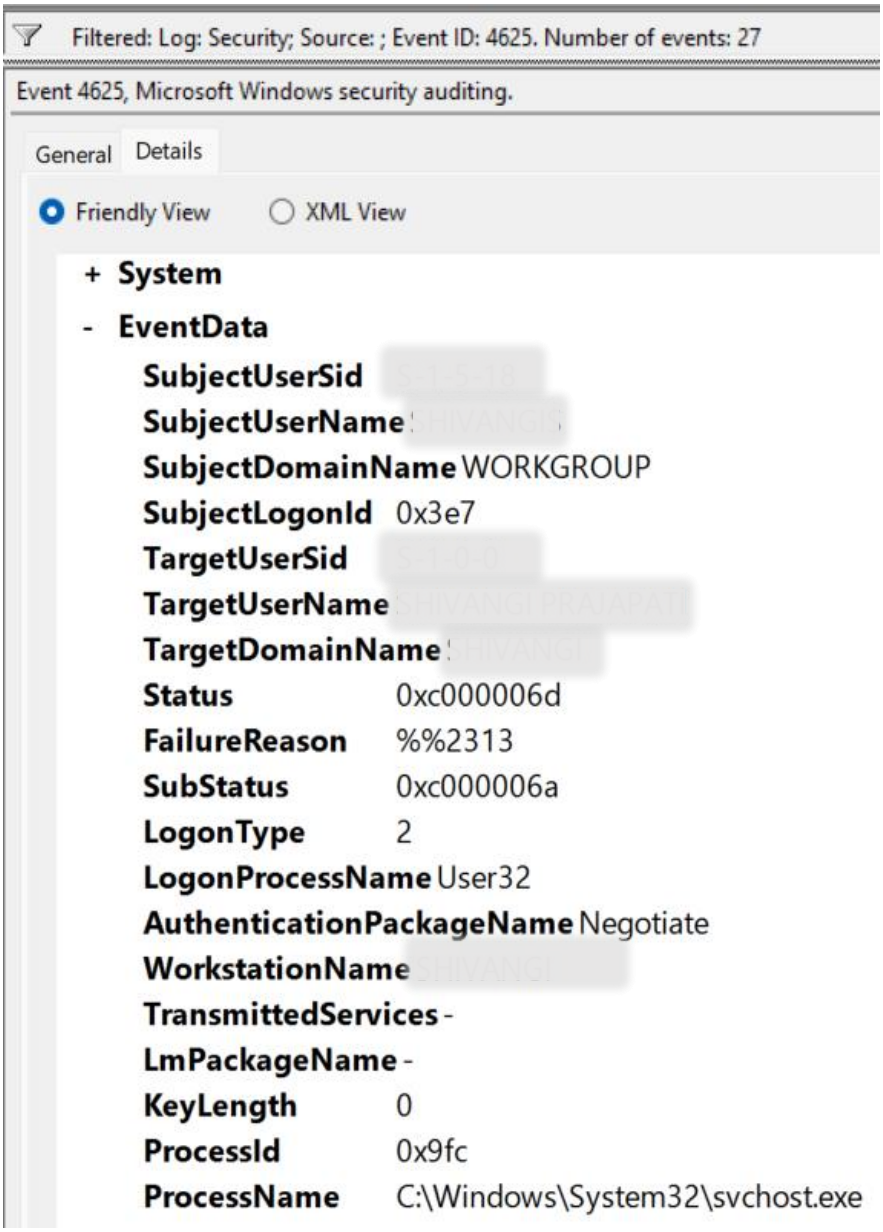
Security Relevance

These events are commonly reviewed Event ID like 4624 when investigating account compromise, unauthorized access, or suspicious authentication activity.

5. Event ID 4625 – Failed Logon

Event ID 4625 records failed authentication attempts and is commonly associated with authentication errors or unauthorized access attempts.





Observation

Several failed authentication events were identified after filtering the Security log. These events recorded unsuccessful attempts to authenticate to the system.

Key Fields Observed

- Target Account Name
- Failure Reason
- Status Code
- Sub Status Code
- Logon Type
- Source Network Address
- Process Information

Analysis

Failed logon events may occur because of incorrect credentials, account lockouts, expired passwords, or authentication configuration issues. While occasional failures

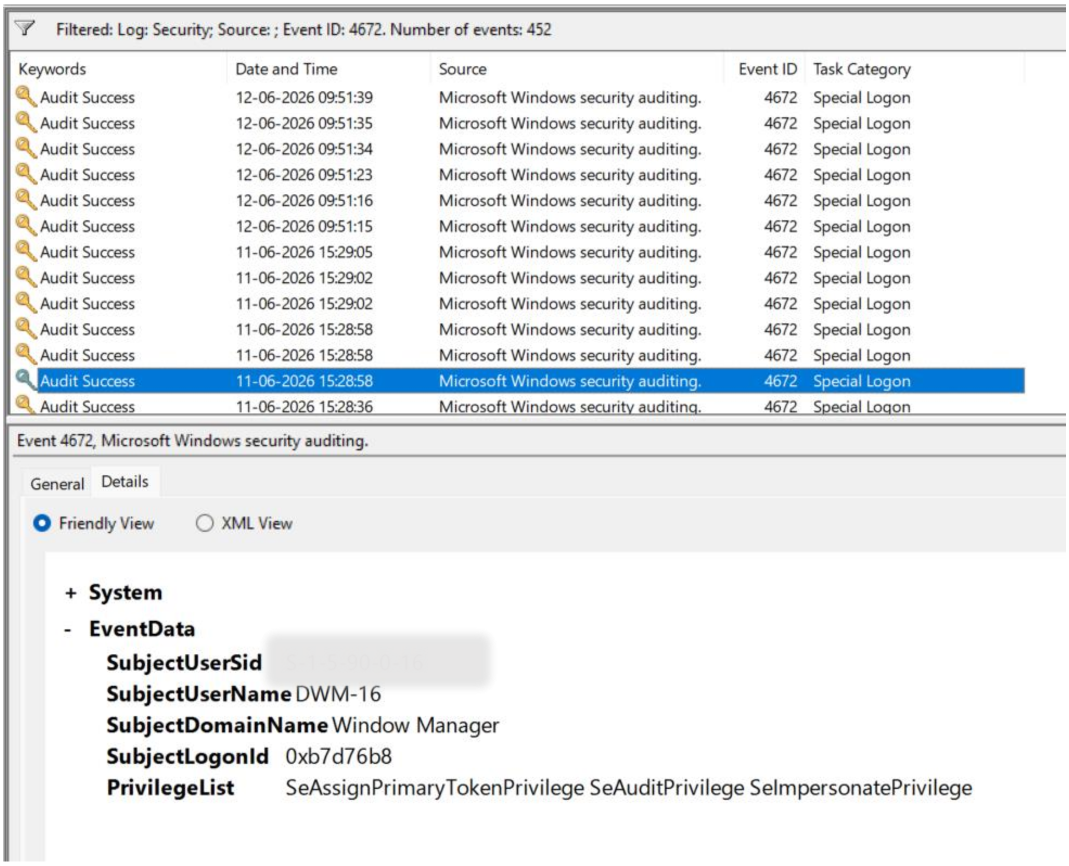
are normal, repeated failed logons can indicate brute-force attacks, password spraying, or unauthorized access attempts.
Reviewing source information and failure codes helps determine whether the activity is legitimate or potentially malicious.

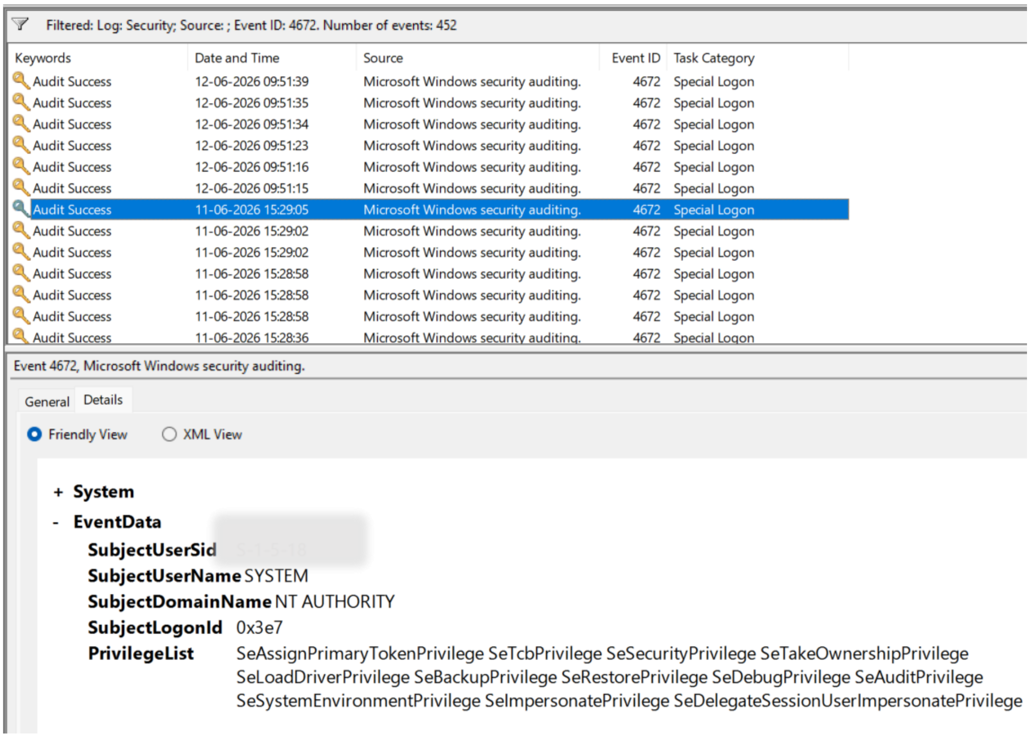
Security Relevance

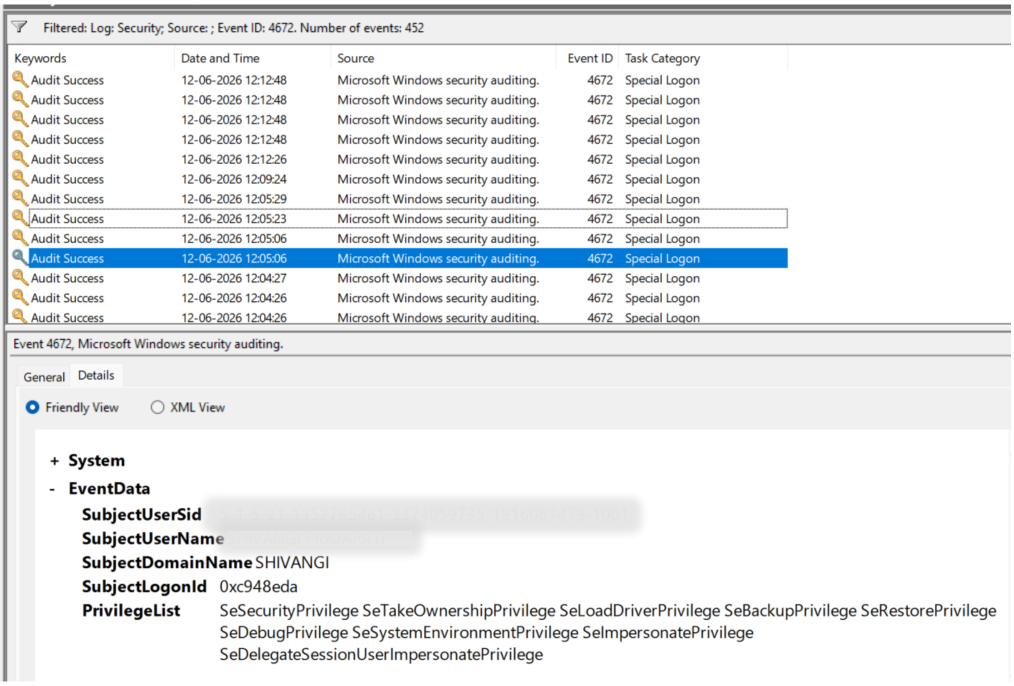
Event ID 4625 is frequently monitored because it can provide early indicators of credential-based attacks.
Failed authentication events provide valuable insight into both user mistakes and potential attack activity targeting user accounts.

6. Event ID 4672 – Special Privileges Assigned

Event ID 4672 records the assignment of elevated privileges during the logon process.







Observation

The filtered results showed multiple Event ID 4672 entries generated during privileged authentication activity.

Key Fields Observed

- Subject User Name
- Subject Domain
- Logon ID
- Privileges List

Analysis

The event details listed privileged rights assigned during authentication. These privileges may include administrative capabilities such as system management, security auditing, backup operations, and debugging privileges.

Such events commonly occur when administrative accounts or Windows services authenticate. Reviewing privileged activity is important because elevated permissions can be abused to perform sensitive actions on a system.

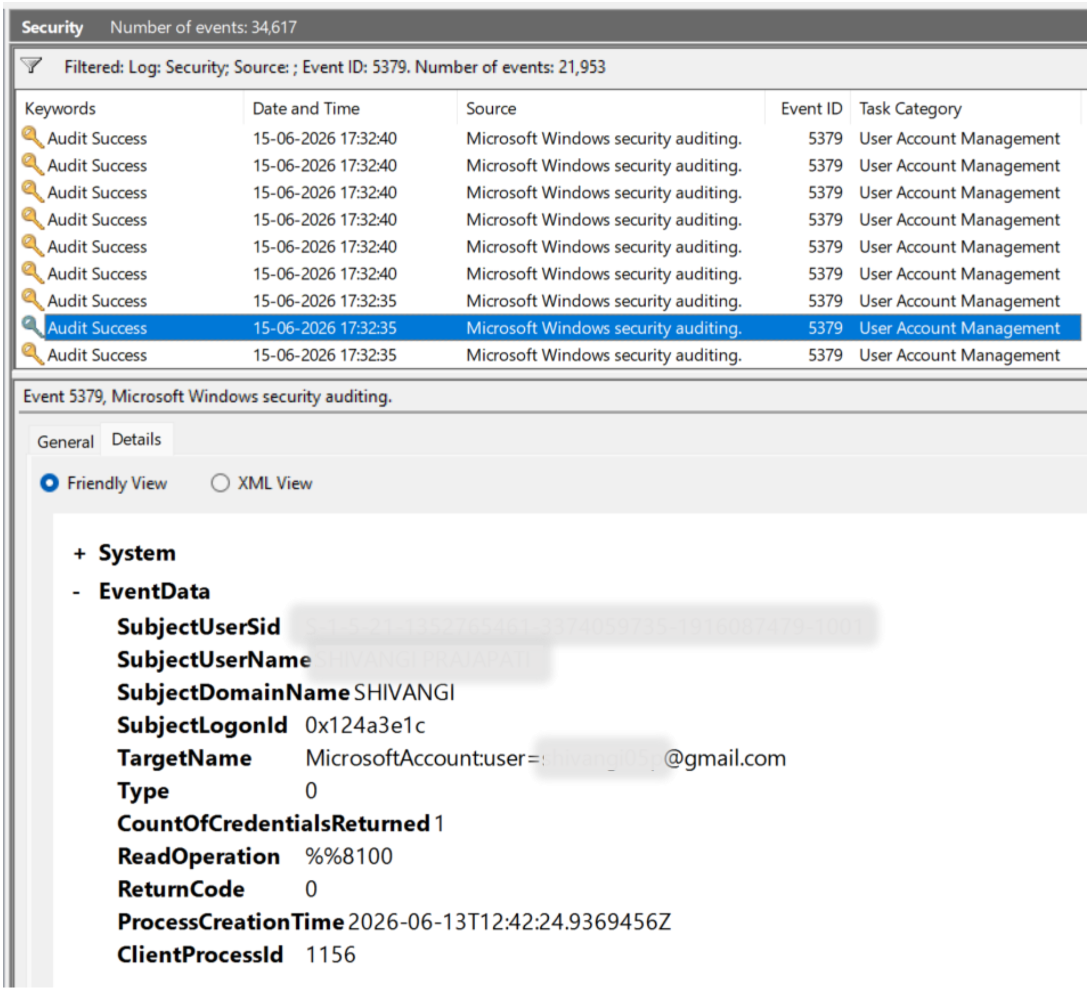
Security Relevance

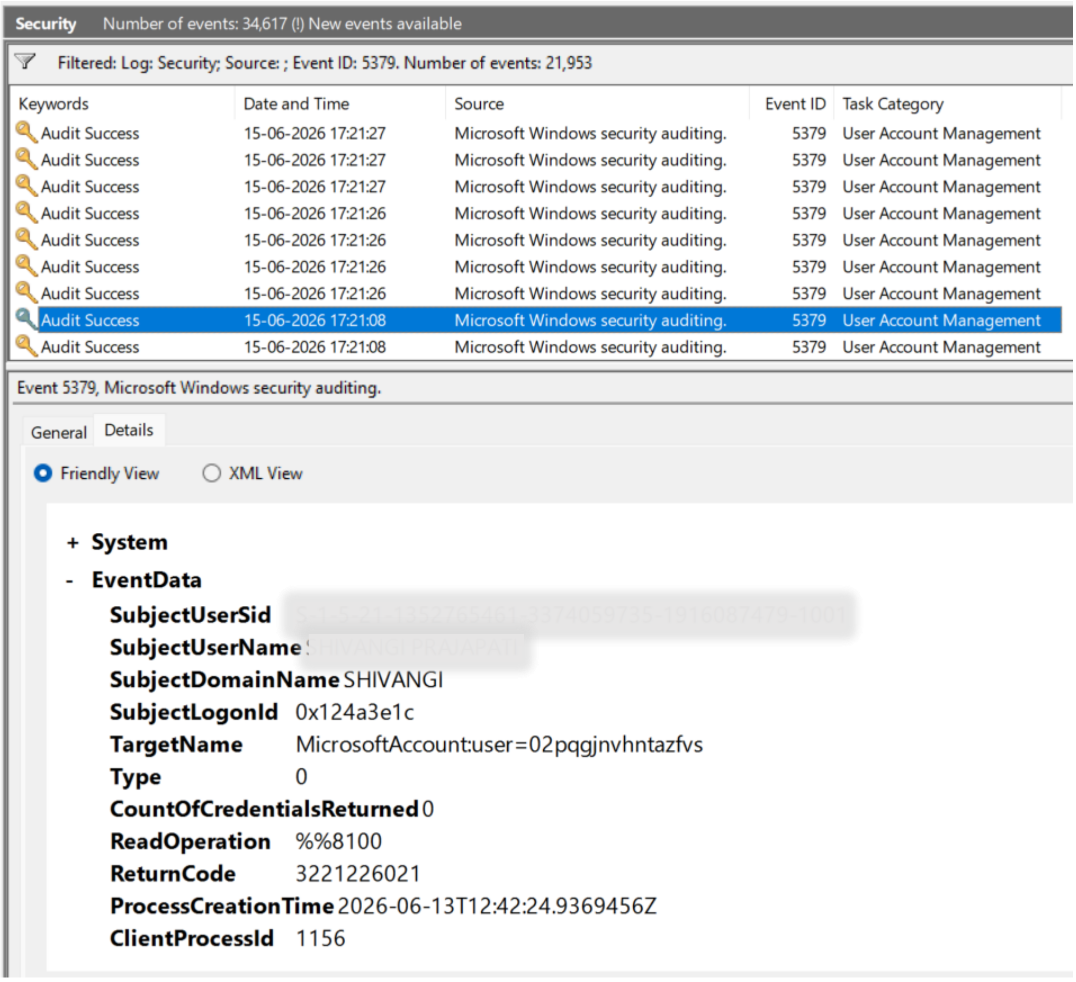
Monitoring Event ID 4672 helps identify privileged account usage and supports investigations involving privilege escalation or administrative account compromise.

Event ID 4672 provides visibility into sessions operating with elevated permissions and highlights potentially sensitive account activity.

7. Event ID 5379 – Credential Manager Access

Event ID 5379 records activity involving credentials stored within Windows Credential Manager.





Observation

A large number of Event ID 5379 entries were identified during analysis, making it one of the most frequently observed events within the Security log.

Key Fields Observed

- Subject User Name
- Target Name
- Credential Type
- Read Operation Information
- Process Information

Analysis

The event indicated that stored credentials were being accessed by applications or Windows processes. Credential Manager is commonly used to store authentication information for websites, network resources, and applications.

Although these events are often generated through normal operating system activity, they remain important from a security perspective because attackers frequently attempt to access stored credentials after gaining initial access to a system.

The high volume of Event ID 5379 events observed during this investigation demonstrates how frequently credential-related operations occur during normal system usage.

Security Relevance

Credential access events provide valuable insight into how authentication information is used across the operating system and may help identify unusual credential retrieval behavior.

Event ID 5379 helps analysts monitor credential usage and investigate potential credential theft or harvesting activities.

8. Event Correlation

Analysis of the Security log revealed several categories of security-related activity:

- Successful authentication events
- Failed authentication attempts
- Privileged account logons
- Credential Manager access events

  Reviewing these events together provides a broader understanding of user activity and authentication behavior within the system.

9. Findings

The investigation produced the following observations:

1. Windows Security logs successfully captured authentication-related activities.
2. Event ID 4624 recorded successful logon events.
3. Event ID 4625 recorded failed authentication attempts.
4. Event ID 4672 identified privileged account activity.
5. Event ID 5379 was one of the most frequently observed events during analysis.
6. Event Viewer filtering simplified the process of locating and reviewing specific event categories.

10. Security Relevance

Windows Security logs are a critical source of information for security monitoring and incident response. These logs assist analysts by:

- Tracking user authentication activity
- Identifying failed login attempts
- Monitoring privileged account usage
- Reviewing credential access activity
- Supporting threat hunting investigations
- Building timelines during incident response

11. Conclusion

This investigation demonstrated how Windows Security logs can be used to analyze authentication and credential-related activities within a system. Through the examination of Event IDs 4624, 4625, 4672, and 5379, valuable insight was gained into user authentication, privileged account activity, and credential usage. The exercise highlighted the importance of Security logs as a primary source of evidence during monitoring and security investigations.